

# Sprint 1 Questions & Assumptions — Instructor Responses

*Software Engineering Practicum CS 5389/5398 — EMR Project*

Thank you for the thorough set of assumptions and questions. Below are answers to guide Sprint 1 development, organized in the same order as your original document.

## Assumptions

- All patient data, notes, and files in the system will be made up, not real patient information.
- This is being built as a learning prototype, not a fully certified clinical system.
- The system will need some way to stop two people from editing the same record at the same time.
- Once a note or record is signed off, it will not be deleted outright — history will be kept and corrections added instead.
- A basic log of who viewed or changed a record and when will be kept.
- The system will use free or low-cost tools since it's a student project with a limited budget.
- The design should make it easy to add more roles, note types, or features later on.
- Students will use one account across their classes/courses.
- After selecting a class, students will see patient cases assigned to that class.
- Teachers and front desk staff are allowed to create new patient files.
- The system will be exclusive to specified UTEP faculty and students.
- The system's admin will have access to user information (students, assigned roles, access levels).

## Critical Workflow & User Stories

**Q: Can you walk us through the step-by-step workflow that your most important user/role follows from beginning to end?**

**A:** The student (learner acting as the provider role) is the most important user in this training/education EHR, so the student workflow below is the anchor workflow — the instructor workflow exists primarily to configure, supervise, and evaluate what the student does.

### Student workflow, start to finish

1. Authentication and role assignment — Student logs in with individual credentials and is placed into the student sub-role of "Provider," scoped to their assigned course/section/cohort. Access is limited to training/test patient records only, never live PHI. - **Follow – Real patients on DB?**
2. Orientation to the assigned case — Student opens an instructor-assigned scenario (a flagged test/training patient with a defined starting status, e.g., "scheduled for visit" or "inpatient/admitted") and reviews the pre-loaded chart: demographics, history, prior notes, medication list, results.
3. Patient status progression — Student advances the encounter through visit-workflow states (checked in → roomed → in progress), reinforcing the status model.
4. Data gathering and clinical reasoning — Student documents subjective/objective findings, reviews existing data, and formulates an assessment using the discipline-appropriate template (SOAP note, medication review, PT/OT eval, etc.).
5. Order entry and documentation — **Student enters mock orders (labs, imaging, medications, referrals) and authors notes within their author-rights scope.** Actions requiring elevated authority (prescribing, controlled substances) are simulated and **routed for co-signature** rather than finalized.
6. Submission for review / co-signature — Student signs or submits the note and orders, which changes the note status to "pending review" and routes the work to the supervising instructor. This is the primary hand-off point.
7. Receiving and acting on feedback — Student reviews instructor annotations/feedback, makes corrections or addenda, and resubmits if required.

8. Encounter closure — Once approved, the note is co-signed/finalized, the patient status is advanced to a terminal state (checked out / discharged), and the case is marked complete for that student.
9. Reflection / self-review — Student may view their graded rubric, comments, and (if enabled) an exemplar note for comparison.

**Q: Can you walk us through the workflow for a student from beginning to end?**

**A:** See the numbered student workflow above.

## **Instructor workflow, start to finish**

1. Authentication and elevated role — Instructor logs in to an instructor/faculty role with supervisory and limited-configuration rights over their assigned cohort.
2. Case and roster setup — Instructor creates or selects test/training patient scenarios, defines the starting patient status and available data, and **assigns cases to individual students or groups**. They confirm the roster and access scoping.
3. Defining expectations — Instructor sets the rubric, required documentation elements, co-signature requirements, and any deliberate "traps" (e.g., a drug interaction or an abnormal result the student must catch).
4. Monitoring in progress — Instructor views a dashboard of student activity and note/encounter statuses (not started, in progress, pending review, completed), tracking who has submitted.
5. Review and co-signature — Instructor opens submitted work, reviews notes and mock orders, and exercises attestation/co-signature authority — the same hierarchy real attendings use over trainees.
6. Feedback and grading — Instructor annotates the chart, scores against the rubric, and either approves (finalize/co-sign) or returns the work for revision (re-routes to the student).
7. Cohort-level oversight — Instructor reviews aggregate performance and audit logs (who accessed what, timeliness, common errors) to inform teaching, then archives completed cases or resets scenarios for reuse.
8. Closeout — Instructor confirms all students reached a terminal encounter state, releases grades/feedback, and archives or recycles the training patient records.

**Q: Can you walk us through the workflow for an instructor from beginning to end?**

**A:** See the numbered instructor workflow above.

## **Where the student and instructor workflows intersect or hand off**

- Case assignment (instructor → student). The instructor's setup step (creating the scenario and setting the initial patient status) is the trigger that populates the student's assigned-case queue. Nothing appears in the student workflow until this hand-off occurs.
- Rubric/expectations (instructor → student, often implicit). What the instructor defines as required elements silently governs what the student is expected to document; making the rubric visible to the student is a design choice.
- Submission for review (student → instructor). When the student signs/submits, the note status flips to "pending review" and routes to the instructor's review queue. This is the single most important hand-off — the point where authority transfers from author to attestor.
- Co-signature / attestation (shared, instructor-controlled). The student holds author rights but not final-signature authority; the encounter cannot reach a terminal, "official" state until the instructor co-signs. This mirrors the trainee/attending co-signature hierarchy in the access-control design.
- Return-for-revision loop (instructor → student → instructor). If the instructor returns the work, the case re-enters the student's active queue, and the student's resubmission re-triggers the review hand-off. This loop can iterate until approved.
- Terminal status and archiving (student action, instructor confirmation). The student advances the encounter to checked out/discharged, but the instructor's closeout/archive step is what finalizes the record and removes it from the active training panel.
- Audit visibility (asymmetric shared view). Both act within the same training patient record, but the instructor sees the full audit trail of the student's actions, whereas the student sees only their own work — an intentional oversight asymmetry.

*Recommended for the design document: represent this as a two-swimlane diagram (student lane, instructor lane) with four hand-off arrows — assign, submit, co-sign, return-for-revision — drawn between them, since those arrows are exactly the points where role permissions and note statuses change hands.*

**Q: Can you highlight any points where the student and instructor workflows intersect or hand off to each other?**

**A:** See the hand-off points listed above (case assignment, rubric/expectations, submission for review, co-signature/attestation, return-for-revision loop, terminal status & archiving, and audit visibility).

## Roles & Permissions

**Q: Can a user have more than one role?**

**A:** Yes, a user can have more than one role.

**Q: Will students have different roles depending on their major?**

**A:** Yes, roles will differ depending on the student's major.

**Q: Who is allowed to create, sign, amend, or close an encounter or clinical note?**

**A:** This depends on the discipline/major, but generally all clinical roles (e.g., pharmacy, nursing, physical therapy) should be able to create, sign, amend, and close their own encounters or clinical notes, since these disciplines act independently.

**Q: Which roles should we prioritize for the prototype?**

**A:** Prioritize pharmacy first, then physical therapy, occupational therapy, and speech language pathology.

## What specific content belongs in each role for security purposes?

| Role                      | Documentation / Content                                                                                 | Security Scope                                                                                                                                                           |
|---------------------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pharmacy                  | Drug therapy problems, medication reconciliation, MTM/comprehensive medication review notes.            | Provider-level: create, sign, and amend own notes. No admin/biller access.                                                                                               |
| Nursing                   | Nursing assessments, vitals, care plans, medication administration.                                     | Provider-level: create, sign, and amend own notes. No admin/biller access.                                                                                               |
| Physical Therapy          | SOAP treatment notes, ROM/strength/gait measures, goal tracking.                                        | Provider-level: create, sign, and amend own notes independently. No admin/biller access.                                                                                 |
| Occupational Therapy      | OT evaluations and treatment notes, functional goal tracking.                                           | Provider-level: create, sign, and amend own notes. No admin/biller access.                                                                                               |
| Psychology                | Psychological/neuropsych testing, therapy progress notes, treatment plans, risk assessments.            | Full behavioral-health chart read; author rights on psychology notes. Psychotherapy notes kept in a separately access-controlled container.                              |
| Psychiatry                | Psychiatric evaluations, mental status exams, medication management, risk assessments, treatment plans. | Full clinical read/write plus prescribing (including controlled substances/EPCS). Same note-segregation as Psychology.                                                   |
| Speech-Language Pathology | Speech/language/cognitive-communication and swallowing evaluations, plans of care, progress notes.      | Read access to relevant H&P/imaging/nutrition data; authoring rights on SLP notes and diet/communication recommendations.                                                |
| Social Work               | Psychosocial assessments, discharge/transition planning, safety screening, care-planning notes.         | Broad read access for disposition planning; authoring rights on social-work notes; heightened confidentiality where substance-use/behavioral-health content is involved. |

|                      |                                                                    |                                                                                                                                                                                       |
|----------------------|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| In-patient Physician | H&Ps, daily progress notes, admission/discharge summaries, orders. | Broadest access tier: full read/write, full ordering/prescribing (incl. controlled substances), co-signature/attestation authority over trainees.                                     |
| Front Desk           | Scheduling, check-in/out, demographics, insurance intake.          | Read/write on demographic and scheduling fields only — no access to clinical notes, results, or behavioral-health content.                                                            |
| Admin                | User/role management, system configuration, reporting.             | System-administration rights and aggregate/de-identified reporting. Broad clinical chart access avoided unless justified; admin actions are themselves audited.                       |
| Biller               | Charge capture, coding (ICD-10/CPT), claims.                       | Access limited to diagnoses/procedures/encounter data needed to substantiate billing — not the full clinical narrative. Extra restriction on behavioral-health/substance-use details. |

*Cross-cutting design principle: Psychotherapy notes and substance-use records should sit in separately access-controlled containers with break-the-glass logging; "view" and "author" rights should be distinguished per role; sensitive-content access should be fully audited.*

**Q: Are there any additional roles or sub-roles we have not listed?**

**A:** Yes — a main Administrator role, separate from the clinical disciplines, with the ultimate/highest level of access across the system. This admin trains users, assigns permissions based on role, and adds cases or test patients as needed.

**Q: Do you want 'Patient' as a role? (patient portal)**

**A:** Yes — a Patient role should be included, giving patients access to their own medical records.

**Q: What roles should have the 'batch account creation' ability?**

**A:** A dedicated role (working name: "Batch Account Administrator") responsible for creating accounts in bulk.

**Q: What restrictions should be in place for batch account creation?**

**A:** For now, restricted to someone in the Pharmacy role. Since this is for educational purposes only, that person can have access to everything — no additional role-type restrictions, amount limits, or batch-deactivation limits are needed at this stage.

## Platform & Delivery

**Q: Do you plan on using this system on the web or through an application, or both? If both, which takes priority?**

**A:** Ideally both, but web access takes priority — something that can be linked from one of the department's web pages, accessible to faculty, staff, and students.

**Q: Will this system need to connect to or exchange information with any existing systems, such as an EHR, pharmacy, laboratory, or UTEP system? If so, which ones?**

**A:** No — since this is for educational purposes only, it doesn't need to exchange information with real external systems. However, it should have its own landing page, similar to UTEP's existing simulation landing page, to serve as an access point for this educational EMR.

**Q: What parts of your current EMR (EHR Go) work well for you, and why?**

**A:** EHR Go allows an instructor to create a patient case (name, age, labs, history of present illness, etc.) and share it with students via a link for them to work through. This case-building and sharing workflow is useful for teaching.

**Q: What do you like or dislike about how EHR Go handles role-based information and medication administration?**

**A:** The main dislike is that EHR Go doesn't reflect true EMR functionality — in a real system, once a patient record is created, that information is locked/fixed based on what was originally entered, and providers look up and reference the patient's existing record rather than being able to freely change it. EHR Go doesn't fully replicate this real-world behavior.

**Q: After students work on a test patient, should instructors be able to restore that patient to its original state for reuse, or should the students' changes remain saved?**

**A:** There should be multiple test patients (e.g., Test Patient A, B, C) that instructors can assign to students. Instructors should be able to restore a test patient to its original state for reuse, since these are meant purely for students to practice navigating the system. For actual assignments, students are given a real case to develop, and their completed work is sent to the instructor for sign-off. Multiple students working on "the same patient" would each have their own unique copy/instance under their own profile, so their individual work can be independently reviewed and signed off — the exact mechanism for this still needs to be worked out with the dev team.

**Q: Are there any UTEP accessibility requirements that this system needs to follow?**

**A:** As a state institution, the system should follow ADA accessibility standards — including accommodations for vision and hearing (e.g., screen reader compatibility, adjustable text/contrast) — and should support dictation/voice input as well.

**Q: Are there any compliance requirements other than HIPAA that should be considered?**

**A:** HIPAA should still be treated as a training/simulation standard, since it reflects real-world practice students will need to follow. In addition, FERPA needs to be considered since this is an educational tool — particularly around grading, ensuring students are not able to see their own grades within the system.

**Q: Out of all the features, what is the first and most important feature you want to see demonstrated at our next meeting?**

**A:** Medication interaction checking would be a great (though difficult) feature, but the top priority for the next demo is the basic note-creation workflow — how notes get added to a patient record.

## **Classes & Education Use**

**Q: Will instructors assign specific patient cases to individual students?**

**A:** Yes, this is the ideal. The platform should have dual functionality: (1) Training/Practice Mode — students use shared test patients to freely navigate the system and learn core functions (adding medications, notes, etc.), with no grading attached; and (2) Grading/Assessment Mode — a separate mode where the instructor assigns a specific patient case to a student. The case functions like a real medical record, but is isolated per student — no other students' notes appear on it. Once complete, it's submitted directly to the instructor for review/grading.

**Q: Should instructors have the ability to create template patients for reuse?**

**A:** Instructors should have the ability to create templates. It would be especially useful to have common templates available for students to use — these would be note/documentation templates rather than reusable patient templates. Patients themselves should only exist in the system after they've actually been "seen," and students should only have access to information relevant to that specific visit.

**Q: How should students be added to or removed from classes in the system?**

**A:** Once given a list of students, the instructor should be able to bulk-upload students (name, email, university ID/800 number) via an Excel file, which adds them to a course. Students are added with "student" privileges, including prescribing ability within the system. At the end of the semester/course, instructors are responsible for manually removing students from the course.

**Q: What type of tasks/assignments will a student receive? Can you show an example of assignments/patient cases?**

**A:** A student is given a patient case with a chief complaint and subjective information (e.g., from a simulated patient interview), which they build into a note in the medical record. Example scenario: "Albert Einstein, 57-year-old male, history of Type 2 diabetes diagnosed 10 years ago, currently on Metformin 500mg/day, last A1C 10.5%, other labs within normal limits, family history of diabetes, drug allergy to penicillin, also taking vitamins." The student would need to recognize that the diabetes is uncontrolled and add/adjust treatment accordingly.

**Q: How would the student signal that they have completed an assignment, and how should the instructor review it?**

**A:** The note-building interface follows a standard SOAP format (Subjective, Objective, Assessment, Plan). In Assessment, the student selects an ICD-10 code, which populates the assessment, and adds a free-text clinical rationale. In Plan, the student documents next steps (medications, education, follow-up timing, referrals). After completing the note, there is a "Submit for Signature" button. Some disciplines (e.g., Physical Therapy) can sign off on their own notes; but for a student learner, even after they "sign" their own note, it must still be routed to an instructor for review and final approval.

**Q: Please also walk us through how a professor would expect to receive an assignment related to the EMR.**

**A:** Instructors should be able to upload a grading rubric to evaluate the student's SOAP note. Once a student signs off and selects the instructor to route the note to, it appears in that instructor's review queue, showing the complete note (vitals, HPI, surgical history, allergies, medications, etc.). The instructor evaluates whether the student correctly assessed/prioritized the disease state(s) and whether the plan (e.g., drug selection) was appropriate and evidence-based. The grade itself should not be stored/displayed within the EMR portal — it should be recorded in a separate system (e.g., the university's LMS/gradebook), consistent with FERPA.

**Q: Will students or staff need to collaborate in the system, such as sharing patient cases or tasks? If so, how should the work?**

**A:** No — ideally, students should not collaborate on a patient case. Each case is meant to be worked independently, to support individual critical thinking and independent learning.

## Patient Records

**Q: What information should be collected when first creating a patient, and which fields are required versus optional?**

**A:** Patient data collection happens in stages, matching real clinical workflow: (1) Registration — demographics: first name, last name, preferred name, date of birth, sex assigned at birth, gender identity, pronouns, etc. (2) Vitals (Medical Assistant) — blood pressure, oxygen saturation, pulse, weight, height. (3) History-Taking — chief complaint, current medications, family history, surgical history, allergies, and social history (marital status, tobacco/alcohol/drug use). (4) Provider Visit — further history related to the chief complaint, plus a full head-to-toe physical exam (pulmonary, cardiovascular, HEENT, musculoskeletal, neurological, GI, rectal, etc.). (5) Medication Review — how each medication is taken, frequency, adherence/missed doses, and access/affordability issues.

**Q: Besides recent visits, demographics, and labs, should the patient summary also show medications, allergies, diagnoses, vitals, upcoming appointments, or referrals?**

**A:** Yes — the system should include the ability to schedule upcoming appointments, and the ability to create referrals to a specific specialty (so students learn how to write referrals properly). Referrals should be visible to the receiving discipline (e.g., a referral sent to Physical Therapy should be visible to the PT role).

**Q: Will patients only be created by instructors, or should students also have access to patient creation?**

**A:** Only instructors can create patients. Students do not have patient-creation access — instructors assign/provide the patient case for students to look up and work on.

**Q: Can you provide one example note template from any discipline so we can understand the expected structure and level of detail?**

**A:** A general clinical encounter note template was provided (Patient ID, CC, HPI using OLDCARTS, Subjective including PMH/PSH/Meds/Allergies/FH/SH, full ROS, Objective with vitals and physical exam by system, Assessment as a problem list, Plan organized by problem, and Disposition/signature). Two discipline-specific examples were also provided: a Pharmacy MTM/comprehensive medication review note (centered on "drug therapy problems," a reconciled medication table, pharmacist recommendations, and monitoring), and a Physical Therapy daily SOAP note (visit number, ROM/strength/gait measures, interventions performed, and a goals-status table). These illustrate that note structure should be discipline-specific rather than one-size-fits-all.

**Q: What patient statuses might be needed?**

**A:** Status should be tracked across separate fields rather than one dropdown, since a patient can be simultaneously active, checked-in, and inpatient: Account/Lifecycle status (Active, Inactive, Prospective, Discharged from practice, Archived, Test/Training patient, Deceased); Encounter/Visit-workflow status (Scheduled, Checked in, In progress, Checkout/Completed, Documented/Signed/Closed); Care-setting/Clinical status (Outpatient, Inpatient/Admitted, Emergency, Discharged with disposition); and Program/Panel-specific status (Enrolled, Plan of care active/completed, Graduated/Goals

met, Lost to follow-up). Terminal statuses should carry required metadata (date/reason), and test/training patients should be clearly flagged and excluded from reporting.

**Q: Who is authorized to update a patient's status?**

**A:** Instructors.

## **Reporting & Data Visibility**

**Q: For the patient summary interface, would all roles see the same laboratory information and results?**

**A:** Yes, all clinical roles should see the same lab information/results. For the patient-facing summary (patient portal view), it should show a summary of the note, demographic information, lab reports/results, and what medications were prescribed.

**Q: Can you provide an example of how detailed the reports should be, and how they should be structured?**

**A:** The patient-facing report should be a summary of what occurred during the visit. Additionally, a valuable feature would be an e-prescribing notification system: when a prescription is written, the patient receives an alert via phone/email that their prescription was sent to a specific pharmacy. This would require creating dummy/simulated pharmacies that "accept" these electronic prescriptions, routed to standardized/dummy patients as part of the simulation.

## **Security**

**Q: Instead of Duo, would texting/emailing a code that they would need to give be acceptable too?**

**A:** Yes — text or email verification codes are preferred, since Duo would be too complicated/burdensome for this use case.